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REVIEW PAPER

Advanced Materials for Land Based Gas Turbines

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Abstract The gas turbine (Brayton) cycle is a steady flow cycle, wherein the fuel is burnt in the working fluid and the peak temperature directly depends upon the material capabilities of the parts in contact with the hot fluid. In the gas turbine, the combustion and turbine parts are continuously in contact with hot fluid. The higher the firing temperature, higher is the turbine efficiency and output. Therefore, increasing turbine inlet temperature (firing temperature) has been most significant thrust for gas turbines over the past few decades and is continuing in pursuing higher power rating without much increase in the weight or size of the turbine. Firing temperature capability has increased from 800 °C in the first generation gas turbines to 1,600 °C in the latest models of gas turbines. Higher firing temperatures can only be achieved by employing the improved materials for components such as combustor, nozzles, buckets (rotating blades), turbine wheels and spacers. These critical components encounter different operating conditions with reference to temperature, transient loads and environment. The temperature of the hot gas path components (combustor, nozzles and buckets,) of a gas turbine is beyond the capabilities of the materials used in steam turbines thus requiring the use of much superior materials like superalloys, which can withstand severe corrosive/oxidizing environments, high temperatures and stresses. However, for thick section components such as turbine wheels, which require good fracture toughness, low crack growth rate and low coefficient of thermal expansion, alloy steels are extensively used. But the wheels of latest models of gas turbines, operating at very high firing temperatures (around 1,300–1,600 °C), are made of superalloy, which offers a significant improvement in stress rupture, tensile and yield strength and fracture toughness required for the application.

Keywords Gas turbine · Combustor · Buckets · Blades · Superalloys · Investment casting

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1 Introduction

Gas turbines have been used for electricity generation for many years. In the past, their use has been generally limited to generating electricity in periods of peak electricity demand. Gas turbines are ideal for this application as they can be started and stopped quickly enabling them to be brought into service as required to meet energy demand peaks. However, small unit sizes and low thermal efficiency of previous turbines restricted the opportunities of their wider use for electricity generation.

There are two basic types of gas turbines—aeroderivative and industrial. As their name suggests, aeroderivative units are aircraft jet engines modified to drive electrical generators. These units have a maximum output of 40 megawatt (MW). Aeroderivative units can produce full power within 3 min after start up. They are not suitable for base load operation. Industrial gas turbines range in sizes from 50 to 470 MW and up to 680–700 MW in combined cycle. Depending on size, start up can take from 10 to 40 min to produce full output. Over the last two decades there have been major improvements to the sizes and efficiencies of these gas turbines and they have a lower capital cost per kilowatt installed than aeroderivative units and, because of their more robust construction, are suitable

Table 1 Damage mechanisms in gas turbine components

Components	Creep	HCF	LCF	Corrosion	Oxidation	Wear
Combustor	**	**	**	**	**	*
Nozzles	**	**	**	**	**	
Buckets	**	**	**	**	**	
Turbine wheels	*	**		*		
Compressor blades		**		**		*

* Important, ** very important

for base load operation [1, 2]. These advanced gas turbines employ many advanced directionally solidified (DS) and single crystal superalloys for buckets and nozzles with advanced thermal barrier coatings and internal cooling.

Use of DS or single crystal superalloy buckets exhibits further improvement in creep, fatigue and impact strength over equi-axed buckets. As superalloys have become more complex, it has become increasingly difficult to obtain both higher strength levels and satisfactory corrosion resistance at elevated temperatures [3–5]. Correspondingly, the trend towards higher firing temperature increases the need for protective coatings, which almost doubles the component life. To sustain the consistent increase in firing temperature, various improved coatings have been applied. To extend the use of existing material at still higher firing temperatures, efficient cooling methods have been developed for hot gas path components, turbine wheels and spacers etc. to withstand the damages encountered during service. Various damage mechanisms encountered by gas turbine components are given in Table 1.

This paper describes the operational requirements of gas turbine components selection criteria for the materials employed for such applications. Some of the materials used for gas turbine application are given in Table 2 and the chemical composition of the materials used by GE is given in Tables 3 and 4 [6].

2 Gas Turbine Design, Operation and Materials

The design and manufacture of gas turbines for power generation system is specified/regulated by the American Petroleum Institute Standard 616 (small to intermediate engines). Gas turbine thermal efficiency increases with increasing temperature of the gas flow exiting the combustor and entering the work-producing component—the turbine. Turbine entry temperatures (TET) in the gas path of modern high-performance land based gas turbines operate at 1,600 °C or lower. In high-temperature regions of the turbine, special high-melting-point nickel-base superalloy blades and nozzles (vanes), which retain strength and resist hot corrosion at extreme temperatures, are used. These superalloys, when conventionally vacuum cast, soften and melt at temperatures between 1,200 and 1,500 °C. That means blades and nozzles closest to the combustor operate in gas path temperatures far exceeding their melting point and are cooled to acceptable service temperatures (typically eight- to nine-tenths of the melting temperature) to maintain integrity [6–8]. Cross section of a Frame 6 gas turbine is shown in Fig. 1 [9]. Figure 2 shows a four-stage GE turbine, which consists of a significant number of single crystal and DS investment cast parts [10]. Chronological development and evolution of advanced materials for buckets and nozzles is shown below in Table 5.

The following sections describe the current and anticipated component design and operating conditions for the stages of small to intermediate and larger industrial gas turbines and aim to identify the technical challenges and requirements.

2.1 Compressor

For small to intermediate industrial gas turbine (IGT) compressors, the temperatures experienced currently range from −50 to less than 500 °C, and usually do not present any significant challenges to the materials engineers. The continued use of low alloy and ferritic stainless steels has proved to be adequate and this situation is likely to continue unless significant increases in compressor temperatures are needed due to much higher-pressure ratios and rotor speeds. In such a situation it has been assumed that aero-derivative technology such as titanium alloys, nickel alloys, intermetallics and composites will be employed (Sect. 3). This would, however, present a significant increase in cost and manufacturing complexity (forgings, machining, joining, component lifing) as well as operational difficulties (component handling, overhaul, repair, cleaning) and may introduce additional problems associated with thermal mismatch and fretting fatigue from adjoining ferritic alloys [1].

For large utility power generation engines, however, targeting >60 % efficiency and with >500 MW combined cycle gas turbine (CCGT) performance, the temperature and strength limitations of the rotor steels used currently are limiting the achievement of these performance capabilities [11, 12].

Table 2 Advanced materials for various gas turbines components

Components	GE	SIEMENS	ABB	Westinghouse
Buckets	Rene N5 SC Rene N6 SC DS GTD444 GTD111, DS, SC IN738, U500	IN738LC, IN713 DS, SC, IN792 Nim 90, 80A PWA 1483 SC	CMSX2,4, CM247LC IN713LC, IN738LC IN939, U720, 520 Nim 90, 80A	IN738, ×750 DS, CM247LC, MGA1400 DS WES DS, WES SC,
Nozzles	FSX414 DS, SC X 45/40, N155 GTD111, TD222, Rene N5 SC	IN939 DS, SC PWA 1483 SC	IN939, DS CM247LC FSX414 DS, SC X 45/40, MM509	ECY768 (MM509) IN 939 MGA2400 DS & SC WES 100, X 45
Combustors	HS188, Nim263 HASTELLOY-X	15Mo3 with Tiles IN617 Liner		IN617, HS230 & Ni Base with TBC
Turbine rotor	IN718, IN706 M152, A286 CrMoV	X12CrNiMo 1 2		CrMoV
Compressor rotor	CrMoNiV CrMoV	25NiCrMoV 11 5 26NiCrMoV 14 5		
Compressor blades	X10Cr13 CUSTOM 450	X4CrNiMo 16 51X20Cr13 X20CrMo13		

2.2 Combustor

Combustor is the location of the highest gas temperatures, in excess of 3,000 °F (1,650 °C). The thicker sections that occur regularly along both the inner and outer wall contain cooling holes through which compressor discharge air is forced. The convection cooling plus the film of relatively cool air thus formed protect the combustor material from the hot gas. Differences between metal temperature and flame temperature may well exceed 1,500 °F (850 °C). Thermal radiation from the flame to cooler combustor is a significant source of heat. The design objectives for combustor technologies aim to satisfy the commercial requirements by providing reduced costs, reduced emissions (CO_2 and NOx), improved turndown operation, increased life and to meet the demands for new innovative cycles.

The combustors experience the highest gas temperature and are subjected to a combination of high temperature and pressure. Pressure variations in the combustion process can lead to high cycle fatigue, while start-up and shut-down can cause thermal fatigue, emphasizing the requirements for endurance under creep and thermal fatigue. The microstructural changes occurring due to creep, high cycle and cracks due to thermal fatigue can be observed in Figs. 3 and 4. The materials used to counter such problems presently are generally wrought, sheet-formed nickel-base superalloys, such as Hastelloy X, Nimonic 263, Haynes 188 or Haynes 230. These provide excellent thermo-mechanical fatigue, creep and oxidation resistance for static parts and are formable in fairly complex shapes such as combustor barrels and transition ducts. Of equal importance is their weldability, enabling design flexibility and the potential for successive repair and overhaul operations, which is crucial to reducing life-cycle costs [13]. The high thermal loadings imposed often mean that large portions of the combustor hardware need to be protected using thermal barrier coatings. Use of ceramic matrix composites (CMCs) such as SiC fibres in SiC matrix is considered for advanced high efficiency gas turbines proposed to have higher firing temperatures in the range of 1,800 °C.

2.3 Turbine

Each of the turbine sections such as nozzles, blades, turbine discs etc. presents a range of materials and design issues for current and future turbines that are dependent on their size, operation and duty cycle imposed. Evolution of Westinghouse/Mitsubishi turbines with increasing TET and efficiencies is shown in Table 6 [14] and Figs. 5 and 6 [14, 15]. Contribution to output by each component in a gas turbine is shown in Fig. 7 [16].

Table 3 Composition of advanced materials for GTs employed by GE [6]

Component	Nominal composition													
	Materials	Cr	Ni	Co	Fe	W	Mo	Ti	Al	Nb	V	C	B	Ta
Buckets	U500	18.5	BAL	18.5	–	–	4	3	3	–	–	0.07	0.006	–
	RENE 77 (U700)	15	BAL	17	–	–	5.3	3.35	4.25	–	–	0.07	0.02	–
	IN738	16	BAL	8.3	0.2	2.6	1.75	3.4	3.4	0.9	–	0.10	0.001	1.75
	GTD111	14	BAL	9.5	–	3.8	1.5	4.9	3.0	–	–	0.10	0.01	2.8
	GTD444	9.7	BAL	8.0	–	6.0	1.5	3.5	4.2	Nb 0.5	–	0.10		4.7
	Rene N4	9.8	BAL	7.5	–	6.0	1.5	3.5	4.2	Nb 0.5	–		Hf 0.15	4.8
	Rene N5	7	BAL	7.5	–	5.0	1.5	–	6.2		Re 3.0		Hf 0.15	6.5
	Rene N6	4.2	BAL	12.5	–	6.0	1.4	–	5.8	Y0.01	Re 5.4	0.05	B 0.00 4Hf 0.15	7.2
Nozzles (vanes)	X40	25	10	BAL	1	8	–	–	–	–	–	0.50	0.01	–
	X45	25	10	BAL	1	8	–	–	–	–	–	0.25	0.01	–
	FSX414	29	10	BAL	1	7	–	–	–	–	–	0.25	0.01	–
	N155	21	20	20	BAL	2.5	3	–	–	–	–	0.20	–	–
	GTD-222	22.5	BAL	19	–	2.0	–	2.3	1.2	0.8	–	0.10	0.008	1.00
Turbine wheels	IN-706	16	BAL	–	37.0	–	–	1.8	–	2.9	–	0.06	0.006	–
	Cr–Mo-V	1	0.5	–	BAL	–	1.25	–	–	–	0.25	0.30	–	–
	A286	15	25	–	BAL	–	1.2	2	0.3	–	0.25	0.08	0.006	–
	M152	12	2.5	–	BAL	–	1.7	–	–	–	0.3	0.12	–	–
Combustors	309	23	13	–	BAL	–	–	–	–	–	–	0.10	–	–
	HASTX	22	BAL	1.5	1.9	0.7	9	–	–	–	–	0.07	0.005	–
	Nim 263	20	BAL	20	0.4	–	6	2.1	0.4	–	–	0.06	–	–
	HA-188	22	22	BAL	1.5	14.0	–	–	–	–	–	0.05	0.01	–

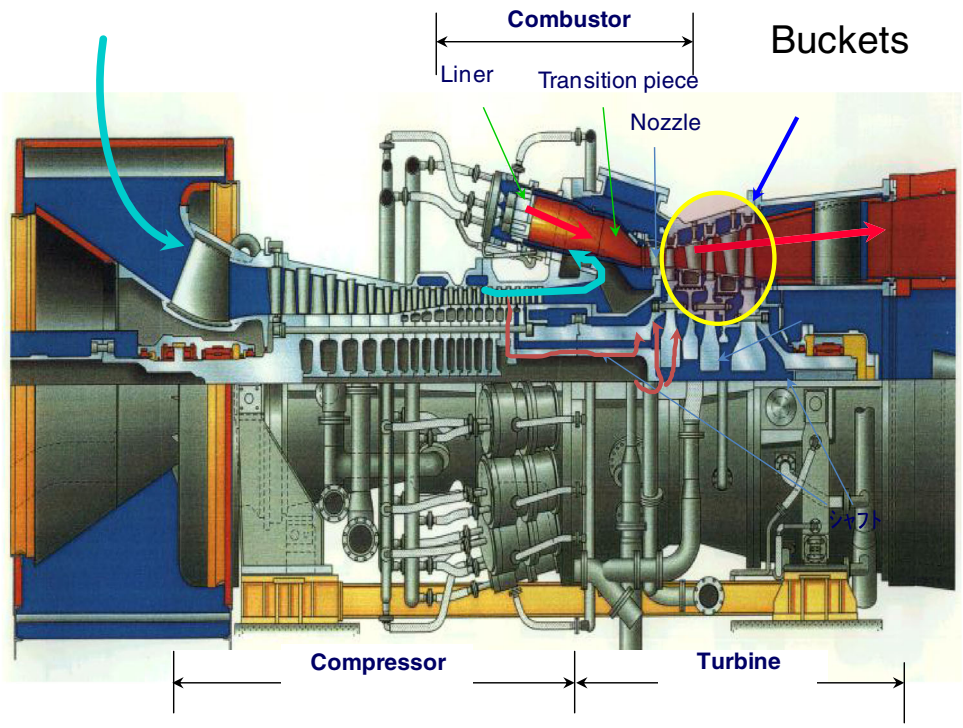


Fig. 1 Cross sectional view of a frame 6 gas turbine [9]

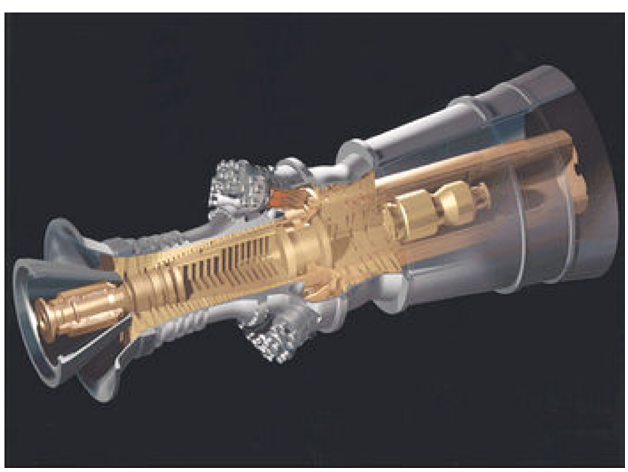


Fig. 2 General electric 'H' 480 MW gas turbine has a rated thermal efficiency of 60 % in combined cycle [10]

2.3.1 Nozzles (Vanes)

Since the gas entering the first stage nozzle can regularly be above the melting temperature of structural metals, cooling is a necessity. Cooling to a uniform temperature over entire nozzle structure is not practical due to a variety of reasons. As a result, temperature differentials can cause thermal stresses that in turn cause low cycle fatigue and fatigue cracking (Figs. 8, 9). Therefore, the nozzle and blade material requirements include corrosion and oxidation resistance or existence of a good protective coating and fatigue and creep strength. Due to continuous long-term operation, precipitation free zone (PFZ) and grain boundary thickening usually occurs in nozzle alloys (Fig. 10a, b).

Material selection for nozzles/vanes is based on alloy strength and material processing as well as requirement from mechanical design and heat transfer consideration. Nozzles/vanes are made from cobalt base superalloys and nickel base superalloys. They are investment cast individually and then welded to a housing to form a nozzle segment or are investment cast as segments. Hence the material must be easily castable into large and complex configurations. A further requirement is weldability for ease of fabrication (cooling inserts are welded in place) and for repair of service induced damage. Alloys used for nozzles typically have greater corrosion resistance but lower creep strength compared to that of blades. FSX 414 is one of the lower strength alloys (Fig. 11) currently used in turbines because it is reported to be readily weldable. Vacuum melted ECY-768 is the latest nozzle material in some designs replacing previously used alloy, X-45, because of its higher creep strength. Vacuum cast alloy Mar-M 509 is also commonly used material in older turbines. ECY-768 alloy is a modified Mar-M 509 with improved castability. MGA2400 has been used in Mitsubishi gas turbines [17–19]. Cast nickel base alloys such as Udimet 500, IN738 and IN939 have also been used for some vanes [20]. However, because it is difficult to produce high quality castings in large multi-vane segments, nickel base alloys have been used for single castings. Large three and four vane segments cast in N155 have also been

Table 4 Composition of advanced materials for GTs employed by Mitsubishi [19]

Component	Nominal composition													
	Materials	Cr	Ni	Co	Fe	W	Mo	Ti	Al	Nb	V	C	B	Ta
Buckets	MGA1400	14	BAL	10	–	4.3	1.5	2.7	4.0	–	–	–	–	4.7
Nozzles(vanes)	MGA2400	19	BAL	19	–	6.0	–	3.7	1.9	1.0	–	–	–	3.5
	Alloy A	23.5	10	BAL	–	7.0	–	0.25	0.20	–	–	–	–	3.5
	Alloy B	25.5	10.5	BAL	–	7.5	–	–	–	–	–	–	–	–
	Alloy C	22.5	BAL	19	–	2.0	–	3.7	1.9	1.0	–	–	–	1.4

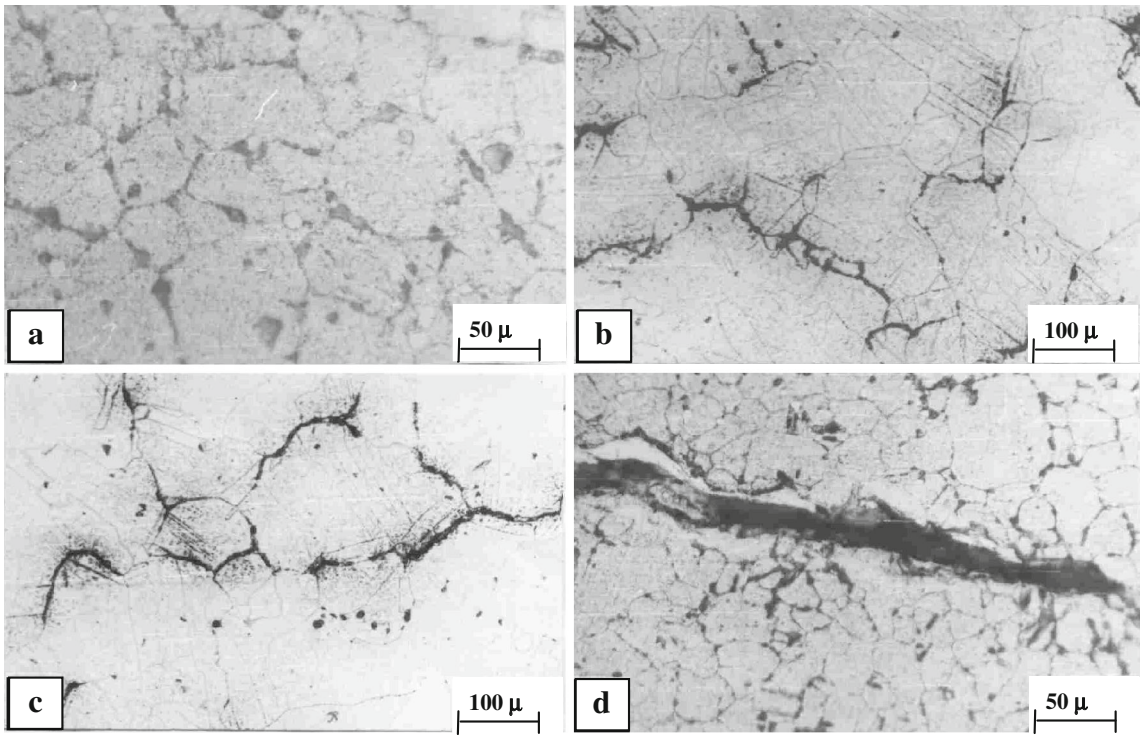


Fig. 3 Microstructure of a combustor of Hastelloy-X showing **a** creep cavities, **b**, **c** micro-cracks and **d** macro-cracks due to creep and fatigue

used for some cooler running last stages (540–650 °C). GTD-222, nickel-based nozzle alloy, was developed by GE in response to the need for improved creep strength in latter stage nozzles such as for stage 2 and 3. It offers an improvement of more than 150 °F/66 °C in creep strength compared to FSX-414, and is weld-repairable. An important additional benefit derived from this alloy is enhanced low-temperature hot corrosion resistance [6, 21].

2.3.2 Turbine Blades

The rotating blades of the turbine convert the kinetic energy of the gas exiting the nozzles to shaft power used to drive the compressor and the load devices. Turbine blades are subjected to significant rotational and gas bending stresses at extremely high temperatures, as well as severe thermo-mechanical loading cycles as a consequence of normal start-up and shut down operation and unexpected trips. The TET for a number of engines is in excess of 1,400 °C, with base metal temperatures ranging from 850 to 1,050 + °C (1,123 to 1,323 + K), depending on the specific turbine type, the cooling efficiency and operation [13]. The target lifetime under these conditions is dependent on turbine type and duty cycle, but can be in excess of 24,000–50,000 operating hours (OH). The blades pass through the combustion gases directed by the combustor and nozzles and are subjected to frequency excitations, which can lead to high cycle fatigue failure. The high-pressure stages are cooled to withstand the hot gas temperatures and, depending on the type of fuel, severe corrosion and erosion of the blade structure can take place. The combination of stress and temperature results in creep

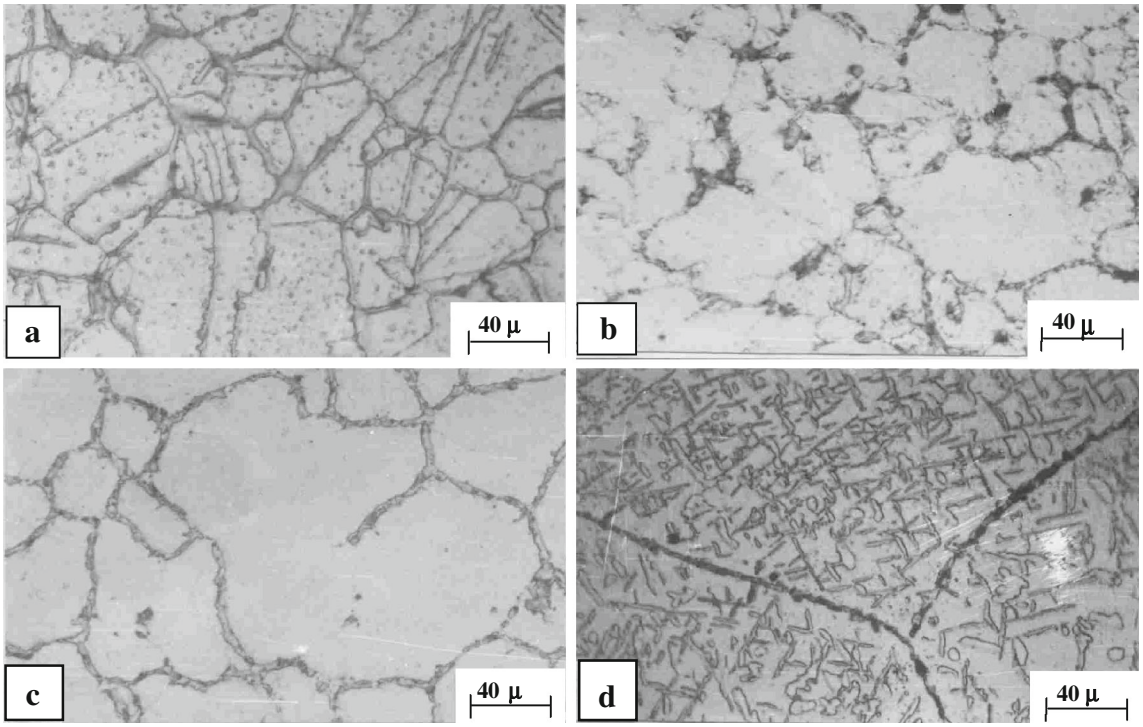


Fig. 4 Microstructure of a transition piece of Hastelloy-X showing **a** healthy microstructure, **b** creep cavities, **c** grain boundary thickening and **d** sigma phase after service exposure

Table 5 Progress in material development for buckets and nozzles in a gas turbine

Components	GT frames	GT stages	Materials	Year of Intdn. as first stage Matl
Buckets	Frame 3, 5, 6, 9	Third stage	U500	1960s
		Second stage	IN738	1970s
		First stage	GTD111	1980s
	Frame 9G & H	Fourth stage	IN738	1970s
		Third stage	GTD111	1980s
		Second stage	GTD111 DS	1990s
		First stage	Rene N5 SC	2000
		First stage	Rene N6 SC	2010
Nozzles	Frame 3, 5, 6, 9	Third stage	N115/GTD222	–
		Second stage	X 45/40	1970s
		First stage	FSX414	1980s
	Frame 9G & H	Fourth stage	GTD222	–
		Third stage	FSX 414	1980s
		Second stage	FSX414 DS	1990s
		First stage	Rene N5 SC	2000

being the primary concern in the design of turbine blades. Blade material selection generally results in the application of an alloy with one of the best creep resistance capabilities. For many years the primary considerations in the design of blades has been to avoid the possibility of creep failure due to the combination of high stresses, temperature and the expected length of running time for land based turbines. This has led to include material requirements such as corrosion and erosion resistance and fatigue and creep strength. Also desirable are tensile strength and toughness. The development of alloys to improve mechanical properties with lower cost (castability, production yields) is a

Table 6 Evolution of the Westinghouse/Mitsubishi gas turbines [14]

Engine	W501A	W501AA	W501B	W501D	W501D5	W701G	W701F5	W701J
First start-up data	1968	1971	1973	1975	1979	1990	2010	2013
Power class (MW)	45	60	80	75	107	255	355	470
TET °F (°C)	1600 (876.5)	1650 (899)	1800 (982)	2000 (1093)	2100 (1149)	2642 (1450)	2732 (1500)	2912 (1600)
Inlet air flow (lb/s)	548	744	746	781	781	–	–	–
Pressure ratio	7.5	10.5	11.2	12.6	14	18	21	23
Thermal efficiency (%)	25	27	30	32	33	38	>40	42
Combined cycle efficiency (%)				52	56	60	62	64

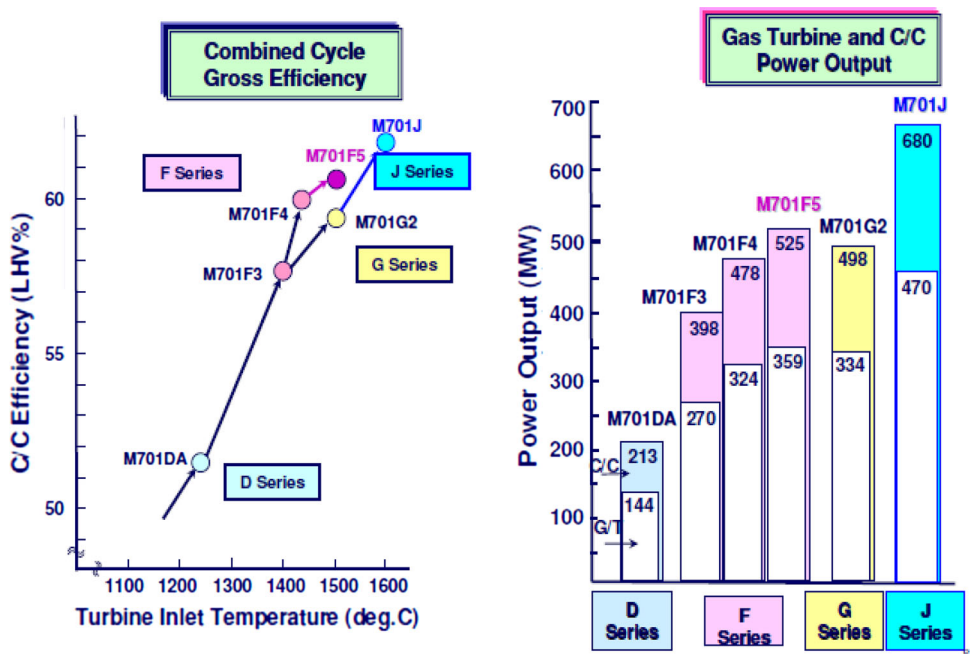


Fig. 5 Performance enhancement of 50 Hz Mitsubishi gas turbines [14]

continuing need and component reliability is of prime importance.

To meet the requirements for increased turbine temperatures, more advanced materials have been introduced into the turbine section of high performance, power generation units. Figure 12 shows a schematic illustration of different temperature loadings to which the first stage blade is exposed for a typical aero and industrial gas turbine [1]. For vanes and blades there has been a gradual move away from conventionally cast nickel-based superalloys, such as IN939, IN738 and IN792 [20], towards DS alloys such as Mar-M247, IN6203DS, GTD111 DS and CM186LCDS [6, 22, 23]. The introduction of these alloys, manufactured using near-net shape investment casting has provided significant benefits in terms of much improved creep and thermal fatigue properties. Further significant benefits have been gained by the use of single crystal (SC) technology using alloys such as CM186LCSX, CMSX-4, GTD111 SC, PWA1484, MGA1400, Rene N5 SC and Rene N6 SC [24]. As a new initiative, a fourth generation single crystal superalloy has been jointly developed by GE, Pratt and Whitney and NASA [3]. This new alloy named EPM102 could provide a 42 °C benefit in creep rupture strength over the second generation blade alloys, PWA1484 and Rene N5 SC [25]. A number of issues are, however, still to be resolved. The increased cost of manufacture, due to high alloying levels and parts rejection, needs to be carefully controlled by the use of revert materials and control of the casting conditions, and offset against improved component lifetimes and more efficient running by enabling higher TET levels to be achieved. To achieve increased creep strength, successively higher levels of alloying additions (Al, Ti, Ta, Re, W) have been used to increase the level of precipitate and substitutional strengthening available at high temperatures. These alloys are extremely creep resistant and have been the key to the success of the aero gas turbine industry and increasingly the land-based sector. However, as the levels of alloying has increased, the

chromium (Cr) additions have had to reduce significantly to offset an increased phase instability problem wherein deleterious phases precipitate out of solution after long-term thermal exposure. These phases led to limited ductility and reduced strength levels. The consequence of having to reduce the Cr level is the significant reduction in the corrosion resistance of the alloys. This has necessitated the development of a series of protective coating systems to meet the range of fuel types used by various operators. These coatings are applied to provide increased component lifetimes, but they often demonstrate low strain to failure properties that can impact upon the thermo-mechanical fatigue endurance. A typical thermal fatigue crack in a turbine blade is shown in Fig. 13 and corrosion attack in Figs. 14 and 15.

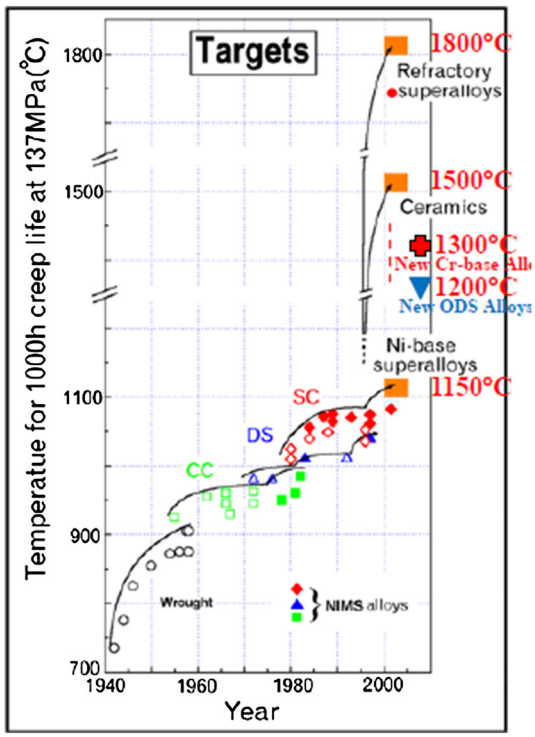


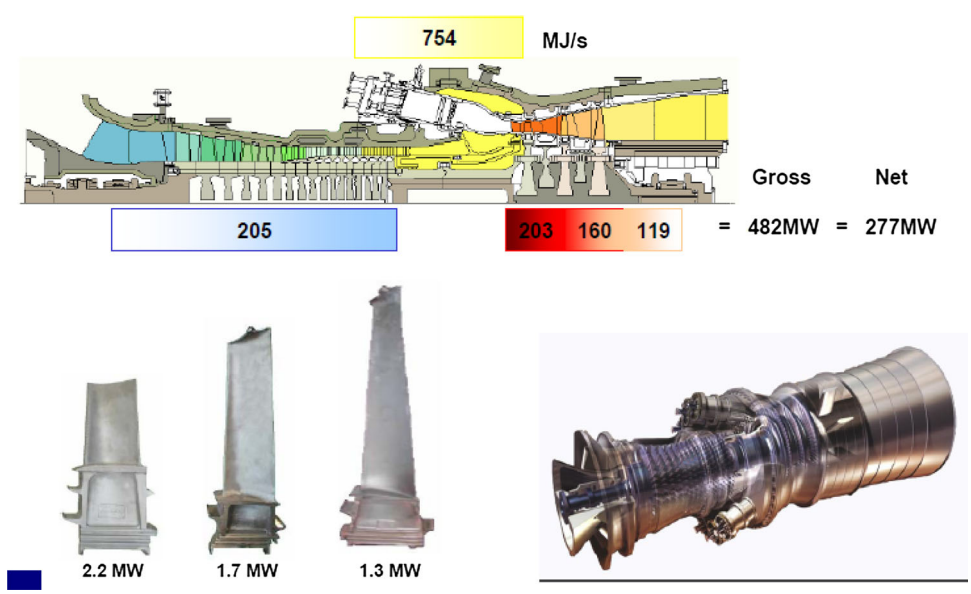
Fig. 6 Chronological development of high temperature materials for gas turbines [15]

The development of IGT-specific turbine blade alloys continues to be a difficult problem to resolve. Much dependence has been placed by the land-based sector on the transfer of advanced technologies from the aero sector and this has not always provided the necessary solutions. The key issues associated with this dependence are as follows:

- Development of a succession of alloys with increasingly lower corrosion resistance despite increasing requirements for the use of differing poor quality fuels and a range of running conditions to satisfy the power generation market requirements.
- Limited castability of large-scale components due to recrystallization and microstructural defects such as freckles, large angle grain boundaries and coarse dendritic structures leading to reduced property levels.

Efforts have been made to address these issues with the development of a number of IGT specific alloys having improved castability, higher corrosion resistance and reduced heat treatment times. Alloys such as SC16, MK4, CMX-11 and SCA425 have been developed with varying degrees of success [1, 4, 25].

Fig. 7 Main components in gas turbine with contribution to output by each component [16]



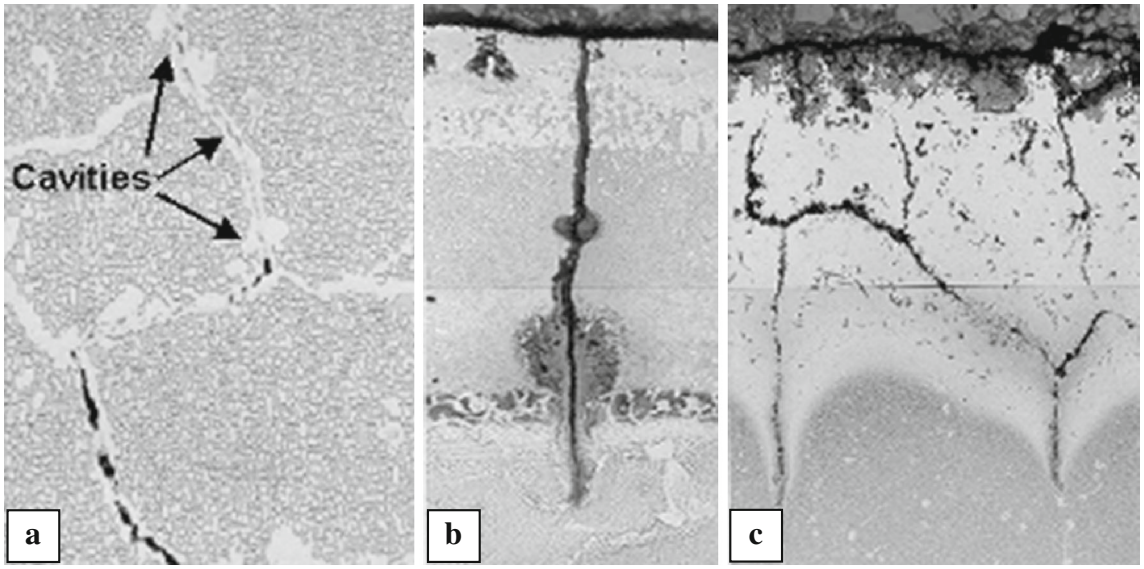


Fig. 8 Micrograph showing **a** creep failure and **b** thermal fatigue failure and **c** oxidation failure

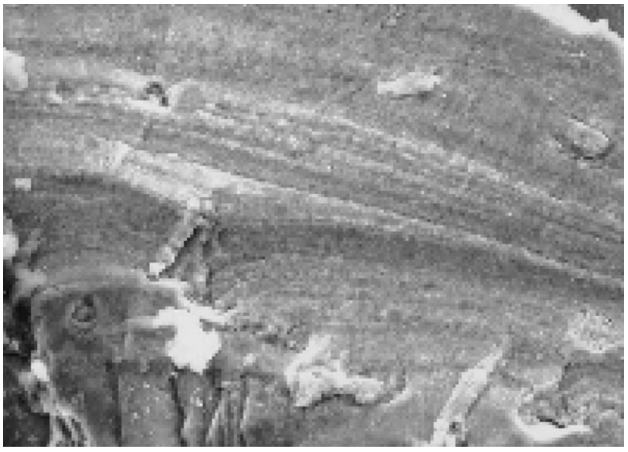


Fig. 9 High cycle fatigue failure

2.3.3 Turbine Discs

The main functions for a turbine disc are to locate the rotor blades within the hot gas path and to transmit the power generated to the drive shaft. To avoid excessive wear, vibration and poor efficiency this must be achieved with great accuracy, while withstanding the thermal, vibrational and centrifugal stresses imposed during operation, as well as axial loadings arising from the blade set, which are attached to the discs by dovetail joints. Under steady-state conditions, turbine disc temperatures can vary from approximately 450 °C in the hub to in excess of 650 °C close to the rim with a requirement for >50,000 h operating life. These temperature loadings are set to increase further across the disc as the demand for improved efficiencies continues. As a practical matter, the temperature is more nearly and not significantly higher than the compressor discharge temperature. Alloy steels are commonly applied in industrial turbines, whereas IN718 and similar alloys are found in aero engines.

Creep and high cycle fatigue resistance are the principal properties controlling turbine disc life and to meet the operational parameters requires high integrity advanced materials having a balance of key properties [1]:

- High stiffness and tensile strength to ensure accurate blade location and resistance to over speed burst failure.
- A combination of high fatigue strength and resistance to crack propagation to prevent crack initiation and subsequent growth during repeated engine cycling.
- Creep strength to avoid distortion and growth at high temperature regions of the disc.
- Resistance to oxidation and corrosion attack and the ability to withstand fretting damage at mechanical fixings.

In order to meet the highest operating temperature and the component stress levels demanded, it has been necessary to develop a series of progressively higher strength steel and Ni-based superalloys, such as IN718, IN706, Waspaloy and U720Li. These are generally manufactured using cast and wrought processing. However, the complex chemistry of these alloys makes production of segregation-free ingots very difficult.

3 Future Developments in Gas Turbine Materials

It is estimated that over the next 20 years a 200 °C increase in turbine entry gas temperature will be required to meet

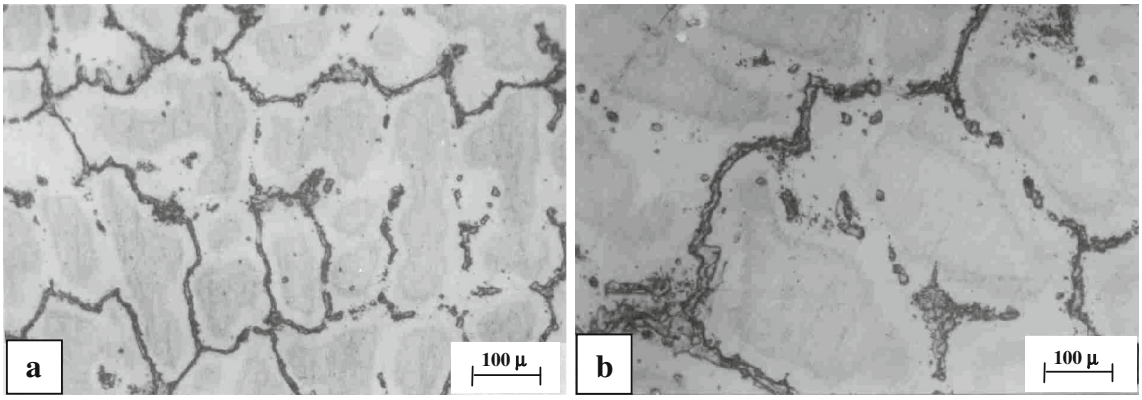


Fig. 10 Microstructure of a nozzle (vane) of cobalt base superalloy showing **a** precipitation free zone near grain boundaries and **b** grain boundary precipitation

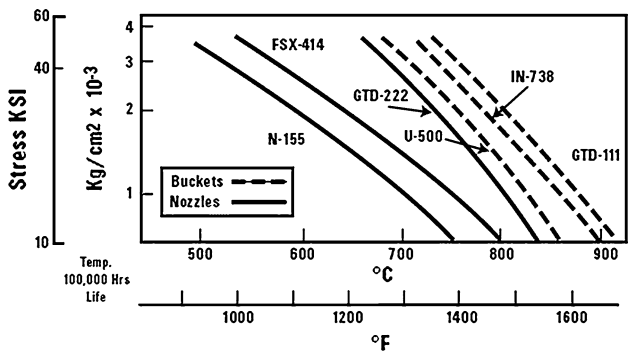


Fig. 11 Comparison of stress rupture properties of blade and nozzle alloys [6]

the demand for improved performance. Some of this increase will be made possible by the further adoption of thermal barrier coatings. These coatings are produced from ceramic pre-cursors and have the potential to contribute about 100 °C through the protection they provide. However, a substantial increase would have to come from the improved design of hot gas path components and use of futuristic materials such as CMCs etc. Lin and Ferber [26] of Oak Ridge National Laboratory, USA have carried out many successful gas turbine field demonstrations using advanced CMCs and proposed high toughness Si_3N_4 ceramic to overcome the issue of foreign object damage (FOD) introduced failure in gas turbines.

3.1 Future Developments in Compressor Materials

The rear end of the high-pressure compressor in an aero-engine is in a temperature environment set by the overall pressure ratio chosen for the engine cycle. Since 1950s, this temperature level has risen by about 300 °C. Titanium alloys have progressively improved in temperature capability up to 630 °C (Fig. 16). This would allow most compressors to be designed completely in titanium. However, practice in the United States has been to switch at approximately 520 °C to nickel alloys and incur a weight penalty.

The development of IMI834 is a good example of the metallurgist's response to the needs of the designer. The requirements were for higher tensile and fatigue strength and enhanced creep performance. These were met by optimizing the structural balance between primary alpha content and the transformed beta phase in the titanium alloy.

Producing integrally bladed discs, or bliscs, is a natural progression in that the blade attachment features are deleted, resulting in significant weight and cost savings. For small engines the most economic manufacturing method is to machine both disc and aerofoils from a single forging. There may be a penalty to pay in that the material strength of the aerofoil may be reduced compared to that of a forged blade. Appropriate attention to the forging method and to the manufacturing processes can overcome this.

3.1.1 Metal Matrix Composites

Titanium metal matrix composites can be applied to both aerofoils and discs. The use of silicon carbide fibre offers about 50 % more strength and twice the stiffness of the high temperature titanium alloys, combined with reduced density. Aerofoil design will benefit from the increased stiffness due to selective reinforcement, providing the ability to control vibration modes and blade untwist. Further exploitation of this technique will be with integrally bladed rings, which are expected to provide a 70 % weight saving relative to a conventional geometry in titanium.

3.1.2 Intermetallics

Another material development program is the use of intermetallics. Compounds of nickel/aluminium and

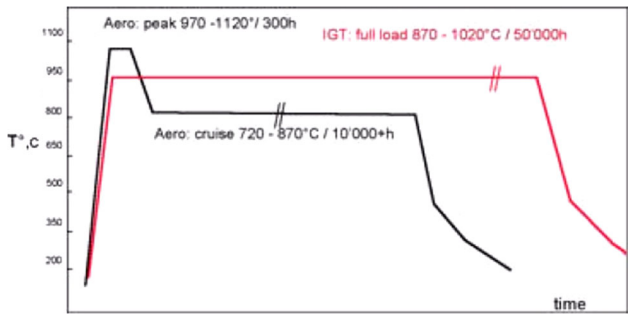


Fig. 12 Schematic illustration of aero and industrial gas turbine temperature loadings [1]

Fig. 13 Photograph of a turbine blade showing thermal fatigue crack on the leading edge

titanium/aluminium have been investigated with current emphasis on the latter. Most intermetallic compounds are brittle at room temperature. The first applications are, therefore, likely to be in small components such as static and rotating compressor aerofoils where the advantages over titanium include higher specific strength and stiffness as well as improved temperature and fire resistance. The use of these materials could extend to more critical components. One possible application is as an alternative matrix to the titanium alloy in a metal matrix composite, although such an application will require alternative fibres to minimize any thermal expansion mismatch and novel processing technology.

3.1.3 Coatings

Some issues associated with rotor corrosion are largely operator dependent, being influenced by compressor washing and cleaning practices and are addressed by protective coatings. Similarly commercially available abradable tip sealing coatings are used to provide and maintain efficiency. Flow path and compressor rotor casings are an effective way to reduce compressor corrosion damage. Two types of barrier and sacrificial coatings are normally

Fig. 14 Photograph of a turbine blade showing corrosion pits

provided [27]. Barrier coatings are overlay coatings applied to flow path surfaces to prevent the contact of corrosive compounds with base material and the sacrificial coatings are also overlay coatings that provide barrier protection as well as interact with corrosive compounds providing corrosion protection to base material. A number of coatings such as electroless nickel, nickel/chromium/cadmium, silicone aluminium and aluminium/ceramic coatings are employed. Flow path coatings help in reduction of flow path deposits thus improving the compressor performance.

Eventually, operating temperatures up to about 800 °C will be possible in the compressors, and intermetallics could offer a very attractive weight saving of around 50 % compared to nickel-based alloys. CMCs of Si_3N_4/SiC combinations are also being attempted. Rolls Royce and Kyocera have carried out field demonstrations of compressor rotors made of Si_3N_4/SiC combinations CMCs [26] for thousands of hours.

3.2 Future Developments in Combustor Materials

Materials technology programmes for future small to intermediate engine combustor designs are aimed at the replacement of conventional wrought nickel-based products with either oxide dispersion strengthened (ODS) metallic systems or CMCs. These programmes are primarily aimed at addressing the limitation in temperature capability and coating compatibility of the conventional alloys used currently. Candidate ODS and CMC materials have been identified and demonstration hardware manufactured and, in the cases of CMC components, engine tested. However, there are limitations to these technologies that need to be addressed. For example, both joining methods (based on for example laser welding or brazing) and coating systems, including TBCs, need to be developed for ODS combustion hardware. These materials have been identified as candidates for efficient, high temperature heat exchangers.

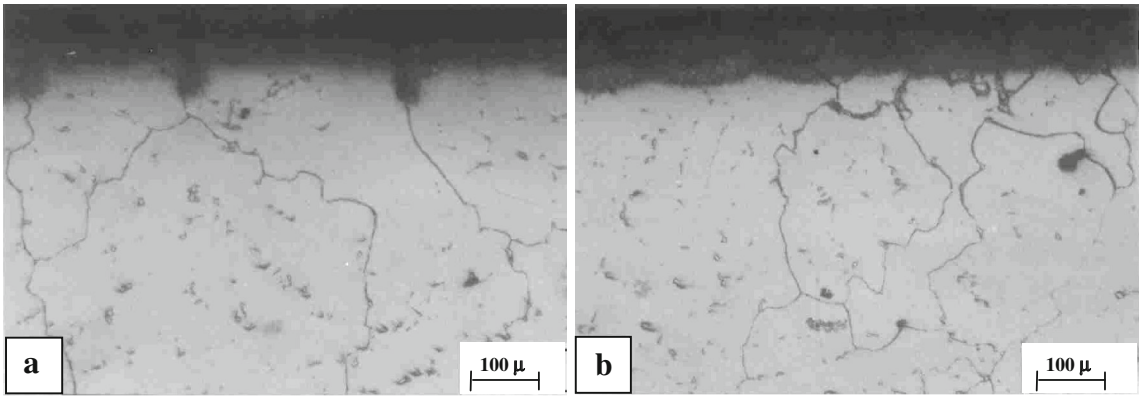


Fig. 15 Microstructure of a turbine blade of IN738 showing corrosion attack

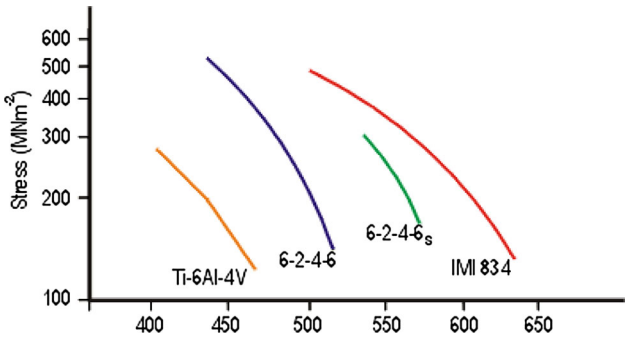


Fig. 16 Progressive improvement of the temperature capability of titanium alloys has reached 630 °C with IMI834

A programme has been under way to establish CMC combustor technology as a viable alternative to metallic systems. However, much work remains to be done to improve the lifetime prediction methods and to develop coatings to provide thermal and environmental protection of the combustor liner. At temperatures in excess of 950 °C, the CMC fibres in SiC/SiC-based composites degrade rapidly as a consequence of oxidation leading to poor structural integrity of the liner during operation. The programme objectives are to develop thermal protection systems (TPS) for these materials that act as a thermal and environmental barrier for the substrate and establish CMC combustor technology running at up to 1,327 °C (1,600 K). Also, there is a need to identify alternative candidate materials based on oxide–oxide CMCs that do not suffer such environmental degradation and potentially offer significant cost savings over the SiC-based materials [28]. GE has tested melt infiltrated SiC/SiC composites (HiPerComp®) and found attractive for high temperature applications in gas turbines as they displayed high thermal conductivity and low matrix porosity [29]. Feasibility of fabricating a wide variety of components has been demonstrated and field engine test of CMC combustor and shroud system for >5,000 h and >10 cycles at shroud material temperature up to ~1,250 °C have been successfully conducted. Caruthers [30] at Oak Ridge National laboratory, USA studied various coatings and their application methods on Si_3N_4 and SiC ceramics composites. Key issues were; coefficient of thermal expansion (CTE) mismatch, mechanical bonding, amorphous to crystalline phase change etc. They tried Yb_2SiO_5 and $MoSiO_2$/mullite environmental barrier coatings (EBC) with Si, and possibly $MoSi_2$ as beneficial bond coatings. During combustion atmosphere exposures, sintering additives remained and concentrated on the surface, providing a possible means of developing protective surface oxides.

3.3 Future Developments in Blade Materials

Untoward phenomena occur at grain boundaries, such as intergranular cavitation, void formation, increased chemical activity, and slippage under stress loading as shown in Fig. 8. These conditions can lead to creep, shorten cyclic strain life, and decrease overall ductility. Corrosion and cracks also start at grain boundaries (Fig. 15). These events initiated at grain boundaries greatly shorten turbine vane and blade life, and lead to lowered turbine temperatures with a concurrent decrease in engine performance. Sufficient understanding of grain boundary phenomena helps in controlling them. In the early 1960s, researchers at jet engine manufacturer Pratt & Whitney set out to deal with the problem by eliminating grain boundaries from turbine airfoils altogether, by inventing techniques to cast single-crystal turbine blades and vanes. The first important development was the DS columnar-grained turbine blade, invented by Frank VerSnyder and patented in 1966. Thereafter many DS and single crystal superalloys have been developed for aero-engine and land based gas turbines, which have revolutionized the concept of TET and opened new vistas for improving gas turbine efficiencies beyond established norms.

For the later turbine stages, such as the low pressure or power turbine, extensive use is made of conventionally cast

alloys such as IN738LC, MarM247 and GTD444 depending on the particular temperature loadings and corrosive environment to be encountered. Recent studies have assessed the potential application of titanium aluminide (TiAl) alloys to meet the needs for harder working, high speed power turbines to provide significant improvements in efficiency (>3 %). Application of TiAl blades would provide much reduced disc stresses, but significant difficulties remain to be resolved that are associated with near-net shape casting, machining and life assessment.

3.3.1 Ceramic Matrix Composites Blades

Further increases in temperature are likely to require the development of CMCs. Today's commercially available ceramic composites employ silicon carbide fibres in a ceramic matrix such as silicon carbide or alumina. These materials are capable of uncooled operation at temperatures up to 1,200 °C, barely beyond the capability of the current best-coated nickel alloy systems. Un-cooled turbine applications will require an all oxide ceramic material system, to ensure the long-term stability at the very highest temperatures in an oxidising atmosphere. An early example of such a system is alumina fibres in an alumina matrix. To realise the ultimate load carrying capabilities at high temperatures, single crystal oxide fibres may be used. Operating temperatures of 1,400 °C are thought possible with these systems. IHI Japan had developed the CMCs for turbine shroud and vane (nozzles) with superior heat resistance properties [31]. Heat-resistant and oxidation-resistant coating have also been developed for these components and rig tests were conducted and confirmed the structural soundness under the TET of 1,923 K (1,650 °C) condition. The CMC vane is made by weaving actual vane shape by combining the airfoil section with shank portion. Tensile and creep strength tests and thermal cycle tests were conducted confirming the manufacturability and sufficient durability of CMC components.

3.4 Future Turbine Disc Materials

To meet the demands for improved technical capability and higher operating efficiencies for small to medium engines, dual alloy and integrally bladed disc (blisc) technologies are being developed. A dual alloy disc enables the differing mechanical property requirements of the hub and rim regions to be reconciled within a single disc structure by combining suitable materials that meet the differing strength–temperature property requirements. This offers considerable advantages over the conventional counterpart in terms of higher temperature and component size capabilities, allowing substantial power and efficiency gains. Recent advances in Europe and in the USA have demonstrated the practicability of joining dissimilar materials to produce small aero engine discs. However, existing knowledge on the success of these joining routes in producing large-scale components and high quality joints is limited by the manufacturing technology. This is currently being developed in conjunction with validated qualification and NDT procedures and lifting methods. Further development and implementation of advanced manufacturing methods will continue to be a high priority for turbine disc applications [28, 32].

4 Summary

The purpose of this paper is an attempt to review some of the materials currently being used in gas turbines and is by no means complete. Major materials development work is ongoing in many laboratories and gas turbine industries to provide a continuous stream of new and improved materials for gas turbine application to meet customers' needs for the most efficient gas turbines. Gas turbine manufacture's intent is to provide the materials necessary for continuously increasing the TET while maintaining the high levels of reliability and availability of the turbine. It is estimated that over the next decades a 200 °C increase in TET will be required to meet the demand for improved performance. This increase will be made possible by the improved design of hot gas path components and use of futuristic materials such as CMCs etc. and further adoption of thermal barrier coatings with more intricate cooling designs in buckets and nozzles.

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